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## Redefining “homework”

It is 6.30 pm on a Tuesday evening and Pragati, a 6th grade student, is just back home from school. In one smooth motion, she flings her school bag to a corner of her one-room home in a slum neighbourhood in Mumbai, and picks up the TV remote to turn on a 17 inch CRT screen. She quickly glances through the program schedule on the direct-to-home service, and chooses the KMN (Kids Movie Network) channel. Realizing that her desired program is to start in 15 minutes, she gets up to wash herself. A mug of water and the drain in front of her door is all she needs to freshen up. Back inside her home, she sits cross-legged, only a couple of feet from the TV screen, and takes out her English notebook to re-read the notes she took in class. She quickly sharpens her pencil, keeps an eraser and a ruler within reach, and unmutes the TV, to watch the opening credits of the latest Peter Jackson masterpiece, “The Hobbit”.

Pragati had already watched the same movie over the weekend, and has mixed feelings about it. But that does not stop her from re-watching it with greater interest, and take notes about its storyline, character development, cinematography, and music score. It helps that the direct-to-home service lets her choose subtitles in her mother tongue, Marathi. Furthermore, having set her age and English proficiency on the set-top box, she is continually shown Marathi and English meanings of words, which an automated algorithm thinks she might find tough to understand.

As the movie comes to an end, Pragati accesses the movie synopsis and director’s comments, via a child-friendly menu. If

she wishes, the service also lets her read the complete screenplay and dialogue. She navigates the cursor over words she does not understand, and opens up a context-sensitive menu and hunts for synonyms and antonyms. Her notes ready, Pragati then navigates to the end-of-movie quiz, enters her school and student ID numbers and her password. Half an hour and 20 questions later, Pragati has finished answering all multiple-choice questions. She gets an additional five minutes to review all her responses, edit a few she is unsure of, and submit. She also gets to choose her favourite questions (“Why did Thoren Oakenshield sing a song at Bilbo’s home?”), which then acts as feedback for the content developers at TVs4Learning, the brains behind TV content-based learning.

She throws both hands up in the air when the instantly calculated score flashes on the screen. An 83% mastery for only the fourth English movie that she has ever watched is no mean feat! She makes a mental note of her marks (to brag about it to his friends at football practice next morning) and reviews the questions that she got incorrect, and the rationale behind their answers. Her homework done, Pragati gets ready for dinner.

## Data is the deity / Keeping our teachers informed

Meanwhile, 3 km away, as Zeenat sits down at her desk and turns on her laptop, she is immediately greeted with the message “You have new assessment data for 35 students!”. A single click opens up a webpage that shows how Pragati and her other students performed on this evening’s movie review assignment. She notes that 95% of the class has scored 90% or more on the factual and vocabulary questions (which brings on a quick smile), but

not so well on inferencing and critical thinking questions. Another click shows her a question-category-wise breakup of scores for each student, while highlighting students that need most attention. After reviewing this data for 10 minutes,

### Masterclass: Knewton (<http://www.knewton.com/>)

Knewton is a US-based technology company that uses data to continuously personalize online learning content for individual students. Knewton analyzes data about the performance of each student and similar students on its proprietary adaptive learning platform, as well as the relevance of the educational content, in order to serve up the best activity for each student at a particular moment in time.

The platform is continuously adaptive, meaning it responds in real time to each student’s activity on the system and adjusts to provide the most relevant content. This means that by the time you get to 9th grade math, it knows what topics in 7th and 8th grade math you didn’t learn properly. You are going to have problems with homework next Tuesday, because you haven’t learnt those concepts properly. Watch this video: <http://dgit.in/eduknewton>